

Quantum Computing and Quantum Machine Learning

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Overview of first week, Basic Notions of Quantum Mechanics

Linear Algebra reminders, Hilbert Spaces, operators on Hilbert spaces, states, qubits and other definitions

1. Mathematical notation, Hilbert spaces and operators
2. Description of quantum systems and one-qubit systems
3. States in Hilbert space, pure and mixed states
4. Operators and simple gates
5. Video of lecture to be added

Reading recommendation: Scherer, Mathematics of Quantum Computations, chapter 2

Practicalities

1. Weekly lectures with weekly exercise sessions and assignments.
The assignments are meant as background for the two projects
2. We plan to work on two projects which will define the content of the course, the format can be agreed upon by the participants but the following topics are those we wish to focus on:
 - ▶ First project: Quantum computing and simulation of quantum mechanical systems
 - ▶ Second project: Continuation of the first topic with more realistic systems and using adaptive VQE
 - ▶ Second project: Applications and implementations of quantum machine learning algorithms
 - ▶ Second project: studies of entanglement and physical realization of quantum gates and circuits

More on projects and final grade

1. Two projects which count 50% each for the final grade
2. Deadline first project March 21, see <https://github.com/CompPhysics/QuantumComputingMachineLearning/tree/gh-pages/doc/Projects/2025/Project1>
3. Deadline second project June 1
4. All info at the GitHub address <https://github.com/CompPhysics/QuantumComputingMachineLearning>

Possible textbooks

1. Maria Schuld and Francesco Petruccione, Machine Learning with Quantum Computers, see <https://link.springer.com/book/10.1007/978-3-030-83098-4>
2. Wolfgang Scherer, Mathematics of Quantum Computing, see <https://link.springer.com/book/10.1007/978-3-030-12358-1>
3. Robert Hundt, Quantum Computing for Programmers, <https://www.cambridge.org/core/books/quantum-computing-for-programmers/BA1C887BE4AC0D0D5653E71FFBEF61C6>
4. Robert Lored, Learn Quantum Computing with Python and IBM Quantum Experience, see <https://github.com/PacktPublishing/Learn-Quantum-Computing-with-Python-and-IBM-Quantum-Exp>

Online material of possible interest

1. See IBM's Quantum Computer Programming: Hands-On Workshop at <https://quantgates.com/learn-quantum>
2. IBM quantum challenge at <https://github.com/qiskit-community/ibm-quantum-challenge-2024?tab=readme-ov-file>
3. IBM learning at <https://learning.quantum.ibm.com/course/>
4. Coding with Qiskit <https://www.youtube.com/watch?v=93-zLTppFZw> and <https://github.com/derek-wang-ibm/coding-with-qiskit/tree/main>
5. Coding with QuTip, see <https://qutip.org/>
6. Qbraid at <https://www.qbraid.com>

Notations and definitions

Vectors, matrices and higher-order tensors are always boldfaced, with vectors given by lower case letter letters and matrices and higher-order tensors given by upper case letters.

Unless otherwise stated, the elements v_i of a vector $\mathbf{v}$ are assumed to be real. That is a vector of length n is defined as $\mathbf{x} \in \mathbb{R}^n$ and if we have a complex vector we have $\mathbf{x} \in \mathbb{C}^n$.

For a matrix of dimension $n \times n$ we have $\mathbf{A} \in \mathbb{R}^{n \times n}$ and the first matrix element starts with row element (row-wise ordering) zero and column element zero.

Some mathematical notations

1. For all/any $\forall$
2. Implies $\implies$
3. Equivalent $\equiv$
4. Real variable $\mathbb{R}$
5. Integer variable $\mathbb{I}$
6. Complex variable $\mathbb{C}$

Vectors

We start by defining a vector $\mathbf{x}$ with n components, with x_0 as our first element, as

$$\mathbf{x} = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \dots \\ \dots \\ x_{n-1} \end{bmatrix} .$$

and its transpose

$$\mathbf{x}^T = [x_0 \quad x_1 \quad x_2 \quad \dots \quad \dots \quad x_{n-1}] ,$$

In case we have a complex vector we define the hermitian conjugate

$$\mathbf{x}^\dagger = [x_0^* \quad x_1^* \quad x_2^* \quad \dots \quad \dots \quad x_{n-1}^*] ,$$

Inner products of vectors

With a given vector $\mathbf{x}$, we define the inner product as

$$\mathbf{x}^T \mathbf{x} = \sum_{i=0}^{n-1} x_i x_i = x_0^2 + x_1^2 + \cdots + x_{n-1}^2,$$

or in case of a complex vector

$$\mathbf{x}^\dagger \mathbf{x} = \sum_{i=0}^{n-1} x_i^* x_i = |x_0|^2 + |x_1|^2 + \cdots + |x_{n-1}|^2,$$

Hermitian conjugate

The hermitian conjugate of a matrix is obtained by taking the complex conjugate of each element and then taking the transpose of the resulting matrix. Often we will just say the transpose or just the conjugate, it should be clear from the context that we will mainly deal with hermitian quantities and our matrices will in most cases be square matrices.

Unitarity, as we will see below, plays also a central role in this course.

Outer products

In addition to inner products between vectors/states, the outer product plays a central role in many applications. It is defined as

$$\mathbf{xy}^T = \begin{bmatrix} x_0y_0 & x_0y_1 & x_0y_2 & \dots & \dots & x_0y_{n-2} & x_0y_{n-1} \\ x_1y_0 & x_1y_1 & x_1y_2 & \dots & \dots & x_1y_{n-2} & x_1y_{n-1} \\ x_2y_0 & x_2y_1 & x_2y_2 & \dots & \dots & x_2y_{n-2} & x_2y_{n-1} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ x_{n-2}y_0 & x_{n-2}y_1 & x_{n-2}y_2 & \dots & \dots & x_{n-2}y_{n-2} & x_{n-2}y_{n-1} \\ x_{n-1}y_0 & x_{n-1}y_1 & x_{n-1}y_2 & \dots & \dots & x_{n-1}y_{n-2} & x_{n-1}y_{n-1} \end{bmatrix}$$

The latter defines also our basic matrix layout.

Basic Matrix Features

A general $n \times n$ matrix is given by

$$\mathbf{A} = \begin{bmatrix} a_{00} & a_{01} & a_{02} & \dots & \dots & a_{0n-2} & a_{0n-1} \\ a_{10} & a_{11} & a_{12} & \dots & \dots & a_{1n-2} & a_{1n-1} \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ a_{n-20} & a_{n-21} & a_{n-22} & \dots & \dots & a_{n-2n-2} & a_{n-2n-1} \\ a_{n-10} & a_{n-11} & a_{n-12} & \dots & \dots & a_{n-1n-2} & a_{n-1n-1} \end{bmatrix},$$

or in terms of its column vectors $\mathbf{a}_i$ as

$$\mathbf{A} = [\mathbf{a}_0 \quad \mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \dots \quad \mathbf{a}_{n-2} \quad \mathbf{a}_{n-1}].$$

We can think of a matrix as a diagram of in general n rows and m columns. In the example here we have a square matrix.

The inverse of a matrix

The inverse of a square matrix (if it exists) is defined by

$$\mathbf{A}^{-1} \cdot \mathbf{A} = \mathbf{I},$$

where $\mathbf{I}$ is the unit matrix.

Selected Matrix Features

Relations	Name	matrix elements
$\mathbf{A} = \mathbf{A}^T$	symmetric	$a_{ij} = a_{ji}$
$\mathbf{A} = (\mathbf{A}^T)^{-1}$	real orthogonal	$\sum_k a_{ik} a_{jk} = \sum_k a_{ki} a_{kj} = \delta_{ij}$
$\mathbf{A} = \mathbf{A}^*$	real matrix	$a_{ij} = a_{ij}^*$
$\mathbf{A} = \mathbf{A}^\dagger$	hermitian	$a_{ij} = a_{ji}^*$
$\mathbf{A} = (\mathbf{A}^\dagger)^{-1}$	unitary	$\sum_k a_{ik} a_{jk}^* = \sum_k a_{ki}^* a_{kj} = \delta_{ij}$

Some famous Matrices

- ▶ Diagonal if $a_{ij} = 0$ for $i \neq j$
- ▶ Upper triangular if $a_{ij} = 0$ for $i > j$
- ▶ Lower triangular if $a_{ij} = 0$ for $i < j$
- ▶ Upper Hessenberg if $a_{ij} = 0$ for $i > j + 1$
- ▶ Lower Hessenberg if $a_{ij} = 0$ for $i < j - 1$
- ▶ Tridiagonal if $a_{ij} = 0$ for $|i - j| > 1$
- ▶ Lower banded with bandwidth p : $a_{ij} = 0$ for $i > j + p$
- ▶ Upper banded with bandwidth p : $a_{ij} = 0$ for $i < j - p$
- ▶ Banded, block upper triangular, block lower triangular....

Matrix Features

Some equivalent statements for square matrices

For an $n \times n$ matrix $\mathbf{A}$ the following properties are all equivalent

- ▶ If the inverse of $\mathbf{A}$ exists, $\mathbf{A}$ is nonsingular.
- ▶ The equation $\mathbf{Ax} = 0$ implies $\mathbf{x} = 0$.
- ▶ The rows of $\mathbf{A}$ form a basis of $\mathbb{C}^n$.
- ▶ The columns of $\mathbf{A}$ form a basis of $\mathbb{C}^n$.
- ▶ $\mathbf{A}$ is a product of elementary matrices. Can you name one example?
- ▶ 0 is not an eigenvalue of $\mathbf{A}$.

Important Mathematical Operations

The basic matrix operations that we will deal with are addition and subtraction

$$\mathbf{A} = \mathbf{B} \pm \mathbf{C} \implies a_{ij} = b_{ij} \pm c_{ij},$$

and scalar-matrix multiplication

$$\mathbf{A} = \gamma \mathbf{B} \implies a_{ij} = \gamma b_{ij}.$$

Vector-matrix and Matrix-matrix multiplication

We have also vector-matrix multiplications

$$\mathbf{y} = \mathbf{Ax} \implies y_i = \sum_{j=0}^{n-1} a_{ij}x_j,$$

and matrix-matrix multiplications

$$\mathbf{A} = \mathbf{BC} \implies a_{ij} = \sum_{k=0}^{n-1} b_{ik}c_{kj},$$

and transpositions of a matrix

$$\mathbf{A} = \mathbf{B}^T \implies a_{ij} = b_{ji}.$$

Important Mathematical Operations

Similarly, important vector operations that we will deal with are addition and subtraction

$$\mathbf{x} = \mathbf{y} \pm \mathbf{z} \implies x_i = y_i \pm z_i,$$

scalar-vector multiplication

$$\mathbf{x} = \gamma \mathbf{y} \implies x_i = \gamma y_i,$$

Other important mathematical operations

and vector-vector multiplication (called Hadamard multiplication)

$$\mathbf{x} = \mathbf{y}\mathbf{z} \implies x_i = y_i z_i.$$

Finally, as already mentioned, the inner or so-called dot product resulting in a constant

$$x = \mathbf{y}^T \mathbf{z} \implies x = \sum_{j=0}^{n-1} y_j z_j,$$

and the outer product, which yields a matrix,

$$\mathbf{A} = \mathbf{y}\mathbf{z}^T \implies a_{ij} = y_i z_j,$$

Defining basis states and quantum mechanical operators

We extend now to quantum mechanics our definitions of vectors, matrices and more.

We start by defining a state vector $\mathbf{x}$ (meant to represent various quantum mechanical degrees of freedom) with n components as

$$\mathbf{x} = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \dots \\ \dots \\ x_{n-1} \end{bmatrix} .$$

Dirac notation

Throughout these notes we will use the so-called Dirac **bra-ket** formalism and we will replace the above standard boldfaced notation for a vector with

$$\mathbf{x} = |x\rangle = \begin{bmatrix} x_0 \\ x_1 \\ x_2 \\ \dots \\ \dots \\ x_{n-1} \end{bmatrix},$$

and

$$\mathbf{x}^\dagger = \langle x| = \begin{bmatrix} x_0^* & x_1^* & x_2^* & \dots & \dots & x_{n-1}^* \end{bmatrix},$$

Inner product in Dirac notation

With a given vector $|x\rangle$, we define the inner product as

$$\langle x|x\rangle = \sum_{i=0}^{n-1} x_i^* x_i = |x_0|^2 + |x_1|^2 + \cdots + |x_{n-1}|^2.$$

For two arbitrary vectors $|x\rangle$ and $|y\rangle$ with the same length, we have the general expression

$$\langle y|x\rangle = \sum_{i=0}^{n-1} y_i^* x_i = y_0^* x_0 + y_1^* x_1 + \cdots + y_{n-1}^* x_{n-1}.$$

The inner product is a real number

Note well that the inner product $\langle x|x \rangle$ is always a real number while for two different vectors $\langle y|x \rangle$ is in general not equal to $\langle x|y \rangle$, as can be seen from the example in the next slide.

We note in bypassing that $|x\rangle^\dagger = \langle x|$, $\langle x|^\dagger = |x\rangle$ and $(|x\rangle^\dagger)^\dagger = |x\rangle$.

Examples

Let us assume that $|x\rangle$ is given by

$$|x\rangle = \begin{bmatrix} 1 - i \\ 2 + i \end{bmatrix}.$$

The inner product gives us

$$\langle x|x\rangle = (1 + i)(1 - i) + (2 - i)(2 + i) = 7,$$

a real number.

Norm

We can use the norm/inner product to normalize the vector $|x\rangle$ and obtain

$$|x\rangle = \frac{1}{\sqrt{7}} \begin{bmatrix} 1 - i \\ 2 + i \end{bmatrix}.$$

As another example, consider the two vectors

$$|x\rangle = \begin{bmatrix} -1 \\ 2i \\ 1 \end{bmatrix},$$

and

$$|y\rangle = \begin{bmatrix} 1 \\ 0i \\ i \end{bmatrix}.$$

We see that the inner products $\langle x|y\rangle = -1 + i$, which is not the same as $\langle y|x\rangle = -1 - i$. This leads to the important rule

$$\langle x|y\rangle^* = \langle y|x\rangle.$$

Outer products

In addition to inner products between vectors/states, the outer product plays a central role in all of quantum mechanics. It is defined as

$$|x\rangle\langle y| = \begin{bmatrix} x_0 y_0^* & x_0 y_1^* & x_0 y_2^* & \dots & \dots & x_0 y_{n-2}^* & x_0 y_{n-1}^* \\ x_1 y_0^* & x_1 y_1^* & x_1 y_2^* & \dots & \dots & x_1 y_{n-2}^* & x_1 y_{n-1}^* \\ x_2 y_0^* & x_2 y_1^* & x_2 y_2^* & \dots & \dots & x_2 y_{n-2}^* & x_2 y_{n-1}^* \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ x_{n-2} y_0^* & x_{n-2} y_1^* & x_{n-2} y_2^* & \dots & \dots & x_{n-2} y_{n-2}^* & x_{n-2} y_{n-1}^* \\ x_{n-1} y_0^* & x_{n-1} y_1^* & x_{n-1} y_2^* & \dots & \dots & x_{n-1} y_{n-2}^* & x_{n-1} y_{n-1}^* \end{bmatrix}$$

Different operators and gates

In quantum computing, the so-called Pauli matrices, and other simple 2×2 matrices, play an important role, ranging from the setup of quantum gates to a rewrite of creation and annihilation operators and other quantum mechanical operators. Let us start with the familiar Pauli matrices and remind ourselves of some of their basic properties.

The Pauli matrices are defined as

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix},$$

$$\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix},$$

and

$$\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Properties of Pauli matrices

It is easy to show that the matrices obey the properties (being involutory)

$$\sigma_x \sigma_x = \sigma_y \sigma_y = \sigma_z \sigma_z = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

that is their products with themselves result in the identity matrix I . Furthermore, the Pauli matrices are unitary matrices meaning that their inverses are equal to their hermitian conjugated matrices. The determinants of the Pauli matrices are all equal to -1 , as can be easily verified.

Commutation relations

The Pauli matrices obey also the following commutation rules

$$[\sigma_x, \sigma_y] = 2i\sigma_z.$$

Before we proceed with other matrices and how they can be used to operate on various quantum mechanical states, let us try to define various basis sets and their pertinent notations. We will often refer to these basis states as our computational basis.

Definition of Computational basis states

Assume we have a two-level system where the two states are represented by the state vectors $|\phi_0\rangle$ and $|\phi_1\rangle$, respectively. These states could represent selected or effective degrees of freedom for either a single particle (fermion or boson) or they could represent effective many-body degrees of freedom. In actual realizations of quantum computing we search often for candidate systems where we can use some low-lying states as computational basis states. But we are not limited to quantum computing. When doing many-body physics, due to the exploding degrees of freedom, we normally search after effective ways by which we can reduce the involved dimensionalities to a number of degrees of freedom we can handle by a given many-body method.

Projection operators

We will now relabel the above two states as two orthogonal and normalized basis (ONB) states

$$|\phi_0\rangle = |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

and

$$|\phi_1\rangle = |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Identity and projection operators

It is straight forward to see that $\langle 1|0\rangle = 0$. With these two states we can define the identity operator I as the sum of the outer products of these two states, namely

$$I = \sum_{i=0}^{i=1} |i\rangle\langle i| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

We can further define the projection operators

$$P = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$

and

$$Q = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Idempotent operators

We note that $P^2 = P$, $Q^2 = Q$ (the operators are idempotent) and that their determinants are zero, meaning in turn that we cannot use these operators for unitary/orthogonal transformations.

However, they play important roles in defining effective Hilbert spaces for many-body studies. Finally, before proceeding we note also that the two matrices commute and we have $\mathbf{PQ} = 0$ and $[\mathbf{P}, \mathbf{Q}] = 0$.

Superposition and more

Using the properties of ONBs we can expand a new state in terms of the above states. These states could also form a basis which is an eigenbasis of a selected Hamiltonian (more of this below).

We define now a new state which is a linear expansion in terms of these computational basis states

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where the coefficients $\alpha = \langle 0|\psi\rangle$ and $\beta = \langle 1|\psi\rangle$ represent the overlaps between the computational basis states and the state $|\psi\rangle$. In quantum speech, we say the state is in a superposition of the states $|0\rangle$ and $|1\rangle$.

Inner products

Computing the inner product of $|\psi\rangle$ we obtain

$$\langle\psi|\psi\rangle = |\alpha|^2\langle 0|0\rangle + |\beta|^2\langle 1|1\rangle = |\alpha|^2 + |\beta|^2 = 1,$$

since the new basis, which is defined in terms of a unitary/orthogonal transformation, preserves the orthogonality and norm of the original computational basis $|0\rangle$ and $|1\rangle$. To see this, consider the unitary transformation (show derivation of preserving orthogonality).

Acting with projection operators

If we now act with the projection operators $\mathbf{P}$ and $\mathbf{Q}$ on the state $|\psi\rangle$ we get

$$\mathbf{P}|\psi\rangle = |0\rangle\langle 0|(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle,$$

that is we **project** out the $|0\rangle$ component of the state $|\psi\rangle$ with the coefficient α while $\mathbf{Q}$ projects out the $|1\rangle$ component with coefficient β as seen from

$$\mathbf{Q}|\psi\rangle = |1\rangle\langle 1|(\alpha|0\rangle + \beta|1\rangle) = \beta|1\rangle.$$

The above results can easily be derived by multiplying the pertinent matrices with the vectors $|0\rangle$ and $|1\rangle$, respectively.

Density matrix

Using the above linear expansion we can now define the density matrix of the state $|\psi\rangle$ as the outer product

$$\rho = |\psi\rangle\langle\psi| = \alpha\alpha^*|0\rangle\langle 0| + \alpha\beta^*|0\rangle\langle 1| + \beta\alpha^*|1\rangle\langle 0| + \beta\beta^*|1\rangle\langle 1|,$$

which leads to

$$\rho = \begin{bmatrix} \alpha\alpha^* & \alpha\beta^* \\ \beta\alpha^* & \beta\beta^* \end{bmatrix}.$$

Finally, we note that the trace of the density matrix is simply given by unity

$$\text{tr}\rho = \alpha\alpha^* + \beta\beta^* = 1.$$

Other important matrices

We present other operators (as matrices) which play an important role in quantum computing, the so-called Hadamard matrix (or gate as we will use later)

$$\mathbf{H} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

The action of the operator $\mathbf{H}$ on a computational basis state like $|0\rangle$ gives

$$\mathbf{H}|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle),$$

and

$$\mathbf{H}|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

that is we create a superposition of the states $|0\rangle$ and $|1\rangle$.

Phase matrix

Another famous operation is the phase matrix given by

$$\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}.$$

Tensor products

Consider now two vectors with length $n = 2$, with elements

$$|x\rangle = \begin{bmatrix} x_0 \\ x_1 \end{bmatrix},$$

and

$$|y\rangle = \begin{bmatrix} y_0 \\ y_1 \end{bmatrix}.$$

The tensor product of these two vectors is defined as

$$|x\rangle \otimes |y\rangle = |xy\rangle = \begin{bmatrix} x_0 y_0 \\ x_0 y_1 \\ x_1 y_0 \\ x_1 y_1 \end{bmatrix},$$

which is now a vector of length 4.

Examples of tensor products

If we now go back to our original one-qubit basis states, we can form the following tensor products

$$|0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = |00\rangle,$$

$$|0\rangle \otimes |1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = |01\rangle.$$

More states

$$|1\rangle \otimes |0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = |10\rangle,$$

and finally

$$|1\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = |11\rangle.$$

Three qubits

We have now four different states and we could try to make a new list by relabeling the states as follows $|00\rangle = |0\rangle$, $|01\rangle = |1\rangle$, $|10\rangle = |2\rangle$, $|11\rangle = |3\rangle$.

In similar ways we can define the tensor product of three qubits (or single-particle states) as

$$|0\rangle \otimes |0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = |000\rangle,$$

which is a new vector of length eight.

Generalizing

We note that with a single-particle basis given the states $|0\rangle$ and $|1\rangle$ we can, with N particles construct 2^N different states. This is something we can generalize to

- ▶ discuss ways of labeling states
- ▶ how to write a code which does it

Tensor products of matrices

The tensor product of two 2×2 matrices $\mathbf{A}$ and $\mathbf{B}$ is given by

$$\mathbf{A} \times \mathbf{B} = \begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix} \otimes \begin{bmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{bmatrix} = \begin{bmatrix} a_{00}b_{00} & a_{00}b_{01} & a_{01}b_{00} & a_{01}b_{01} \\ a_{00}b_{10} & a_{00}b_{11} & a_{01}b_{10} & a_{01}b_{11} \\ a_{10}b_{00} & a_{10}b_{01} & a_{11}b_{00} & a_{11}b_{01} \\ a_{10}b_{10} & a_{10}b_{11} & a_{11}b_{10} & a_{11}b_{11} \end{bmatrix}$$

Measurements

The probability of a measurement on a quantum system giving a certain result is determined by the weight of the relevant basis state in the state vector. After the measurement, the system is in a state that corresponds to the result of the measurement. The operators and gates discussed below are examples of operations we can perform on specific states.

We consider the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Definitions of measurements

1. A measurement can yield only one of the above states, either $|0\rangle$ or $|1\rangle$.
2. The probability of a measurement resulting in $|0\rangle$ is $\alpha^* \alpha = |\alpha|^2$.
3. The probability of a measurement resulting in $|1\rangle$ is $\beta^* \beta = |\beta|^2$.
4. And we note that the sum of the outcomes gives $\alpha^* \alpha + \beta^* \beta = 1$ since the two states are normalized.

After the measurement, the state of the system is the state associated with the result of the measurement.

We have already encountered the projection operators P and Q . Let us now look at other types of operations we can make on qubit states.

Different operators and gates

In quantum computing, the so-called Pauli matrices, and other simple 2×2 matrices, play an important role, ranging from the setup of quantum gates to a rewrite of creation and annihilation operators and other quantum mechanical operators. Let us start with the familiar Pauli matrices and remind ourselves of some of their basic properties.

Assume we operate with σ_x on our basis state $|0\rangle$. This gives

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

that is we switch from $|0\rangle$ to $|1\rangle$ (often called a spin flip operation) and similarly we have

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

More on Pauli matrices

This matrix plays an important role in quantum computing since we can link this with the classical **NOT** operation. If we send in bit 0, the **NOT** gate outputs bit 1 and vice versa. We can use the σ_x matrix to implement the quantum mechanical equivalent of a classical **NOT** gate. If we input what we could represent as bit 0 in terms of the basis state $|0\rangle$, operating on this state results in the state $|1\rangle$, which we in turn can interpret as the classical bit 1.

Linear superposition

If we have a linear superposition of these states we obtain

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}.$$

The σ_y matrix introduces an imaginary sign, which we will later encounter in terms of so-called phase-shift operations.

The σ_z matrix

The σ_z matrix has the following effect

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

and

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix},$$

which we will also link with a specific phase-shift.

Unitarity

The matrices we introduced here are so-called unitary matrices. This is an important element in quantum mechanics since the evolution of a closed quantum system is described by operations involving unitary operations only.

We have defined a new state $|\psi_p\rangle$ as a linear expansion in terms of an orthogonal and normalized basis (our computational basis) ϕ_λ

$$|\psi_i\rangle = \sum_j u_{ij} |\phi_j\rangle. \quad (1)$$

Hamiltonians and basis functions

It is normal to choose a basis defined as the eigenfunctions of parts of the full Hamiltonian. The typical situation consists of the solutions of the one-body part of the Hamiltonian, that is we have

$$\hat{h}_0|\phi_i\rangle = \epsilon_i|\phi_i\rangle.$$

This is normally referred to as a single-particle basis $|\phi_i(\mathbf{r})\rangle$, defined by the quantum numbers i and $\mathbf{r}$.

Unitary transformations

A unitary transformation is important since it keeps the orthogonality. To see this consider first a basis of vectors $\mathbf{v}_i$,

$$\mathbf{v}_i = \begin{bmatrix} v_{i1} \\ \vdots \\ \vdots \\ v_{in} \end{bmatrix}$$

We assume that the basis is orthogonal, that is

$$\mathbf{v}_j^T \mathbf{v}_i = \delta_{ij}.$$

An orthogonal or unitary transformation

$$\mathbf{w}_i = \mathbf{U} \mathbf{v}_i,$$

preserves the dot product and orthogonality since

$$\mathbf{w}_j^T \mathbf{w}_i = (\mathbf{U} \mathbf{v}_j)^T \mathbf{U} \mathbf{v}_i = \mathbf{v}_j^T \mathbf{U}^T \mathbf{U} \mathbf{v}_i = \mathbf{v}_j^T \mathbf{v}_i = \delta_{ij}.$$

Orthogonality preserved

This means that if the coefficients $u_{p\lambda}$ belong to a unitary or orthogonal transformation (using the Dirac bra-ket notation)

$$|\psi_i\rangle = \sum_j u_{ij} |\phi_j\rangle.$$

orthogonality is preserved.

This property is extremely useful when we build up a basis of many-body determinant based states.

Note also that although a basis $\{|\phi_i\rangle\}$ contains an infinity of states, for practical calculations we have always to make some truncations.

Example

Assume we have two one-qubit states represented by

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix},$$

and

$$|\phi\rangle = \gamma|0\rangle + \delta|1\rangle = \begin{bmatrix} \gamma \\ \delta \end{bmatrix}.$$

We assume that the state $|\phi\rangle$ is obtained through a unitary transformation of $|\psi\rangle$ through a matrix $\mathbf{U}$ with its hermitian conjugate $\mathbf{U}^\dagger$ with matrix elements $u_{ij}^\dagger = u_{ji}^*$ and $\mathbf{I} = \mathbf{U}\mathbf{U}^\dagger = \mathbf{U}^\dagger\mathbf{U}$.

Inverse of unitary matrices

Note that this means that the hermitian conjugate of a unitary matrix is equal to its inverse. This has important consequences for what is called reversibility. We say quantum mechanics is a theory which is reversible with a probabilistic determinism. Classical mechanics on the other is reversible in a deterministic way, that is, knowing all initial conditions we can in principle determine the future motion of an object which obey the laws of motion of classical mechanics.

We have then

$$\begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

New basis is also orthogonal

Since our original basis $|\psi\rangle$ is orthogonal and normalized with $|\alpha|^2 + |\beta|^2 = 1$, the new basis is also orthogonal and normalized, as we can see below here.

Since the inverse of a hermitian matrix is equal to its hermitian conjugate/adjoint), unitary transformations are always reversible. Why are only unitary transformations allowed? The key lies in the way the inner product transforms.

To see this we rewrite the new basis from the previous example in its two components as

$$|\phi\rangle_i = \sum_j u_{ij} |\psi\rangle_j,$$

or in terms of a matrix-vector notation we have

$$|\phi\rangle = \mathbf{U}|\psi\rangle,$$

More on orthogonality

We have already assumed that $\langle \psi | \psi \rangle = |\alpha|^2 + |\beta|^2 = 1$.

We have that

$$\langle \phi | i = \sum_j u_{ij}^* \langle \psi | j,$$

or in terms of a matrix-vector notation we have

$$\langle \phi | = \langle \psi | \mathbf{U}^\dagger.$$

Note that the two vectors are row vectors now.

If we stay with this notation we have

$$\langle \phi | \phi \rangle = \langle \psi | \mathbf{U}^\dagger \mathbf{U} | \psi \rangle = \langle \psi | \psi \rangle = 1!$$

Unitary transformations are rotations in state space which preserve the length (the square root of the inner product) of the state vector.

Entanglement

In order to study entanglement and why it is so important for quantum computing, we need to introduce some basic measures and useful quantities. These quantities are the spectral decomposition of hermitian operators, how these are then used to define measurements and how we can define so-called density operators (matrices). These are all quantities which will become very useful when we discuss entanglement and in particular how to quantify it. In order to define these quantities we need first to remind ourselves about some basic linear algebra properties of hermitian operators and matrices.

Basic properties of hermitian operators

The operators we typically encounter in quantum mechanical studies are

1. Hermitian (self-adjoint) meaning that for example the elements of a Hermitian matrix $\mathbf{U}$ obey $u_{ij} = u_{ji}^*$.
2. Unitary $\mathbf{U}\mathbf{U}^\dagger = \mathbf{U}^\dagger\mathbf{U} = \mathbf{I}$, where $\mathbf{I}$ is the unit matrix
3. The operator $\mathbf{U}$ and its self-adjoint commute (often labeled as normal operators), that is $[\mathbf{U}, \mathbf{U}^\dagger] = 0$. An operator is **normal** if and only if it is diagonalizable. A Hermitian operator is normal.

Unitary operators in a Hilbert space preserve the norm and orthogonality. If $\mathbf{U}$ is a unitary operator acting on a state $|\psi_j\rangle$, the action of

$$|\phi_i\rangle = \mathbf{U}|\psi_j\rangle,$$

preserves both the norm and orthogonality, that is $\langle\phi_i|\phi_j\rangle = \langle\psi_i|\psi_j\rangle = \delta_{ij}$, as discussed earlier.

The Pauli matrices again

As example, consider the Pauli matrix σ_x . We have already seen that this matrix is a unitary matrix. Consider then an orthogonal and normalized basis $|0\rangle^\dagger = [1\&0]$ and $|1\rangle^\dagger = [0\&1]$ and a state which is a linear superposition of these two basis states

$$|\psi_a\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle.$$

A new state $|\psi_b\rangle$ is given by

$$|\psi_b\rangle = \sigma_x|\psi_a\rangle = \alpha_0|1\rangle + \alpha_1|0\rangle.$$

Spectral Decomposition

An important technicality which we will use in the discussion of density matrices, entanglement, quantum entropies and other properties is the so-called spectral decomposition of an operator. Let $|\psi\rangle$ be a vector in a Hilbert space of dimension n and a hermitian operator $\mathbf{A}$ defined in this space. Assume $|\psi\rangle$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ , that is

$$\mathbf{A}|\psi\rangle = \lambda|\psi\rangle = \lambda\mathbf{I}|\psi\rangle,$$

where we used $\mathbf{I}|\psi\rangle = 1|\psi\rangle$. Subtracting the right hand side from the left hand side gives

$$[\mathbf{A} - \lambda\mathbf{I}]|\psi\rangle = 0,$$

which has a nontrivial solution only if the determinant $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$.

ONB again and again

We define now an orthonormal basis $|i\rangle = \{|0\rangle, |1\rangle, \dots, |n-1\rangle$ in the same Hilbert space. We will assume that this basis is an eigenbasis of $\mathbf{A}$ with eigenvalues λ_i

We expand a new vector using this eigenbasis of $\mathbf{A}$

$$|\psi\rangle = \sum_{i=0}^{n-1} \alpha_i |i\rangle,$$

with the normalization condition $\sum_{i=0}^{n-1} |\alpha_i|^2 = 1$. Acting with $\mathbf{A}$ on this new state results in

$$\mathbf{A}|\psi\rangle = \sum_{i=0}^{n-1} \alpha_i \mathbf{A}|i\rangle = \sum_{i=0}^{n-1} \alpha_i \lambda_i |i\rangle.$$

Projection operators

If we then use that the outer product of any state with itself defines a projection operator we have the projection operators

$$\mathbf{P}_\psi = |\psi\rangle\langle\psi|,$$

and

$$\mathbf{P}_j = |j\rangle\langle j|,$$

we have that

$$\mathbf{P}_j|\psi\rangle = |j\rangle\langle j| \sum_{i=0}^{n-1} \alpha_i |i\rangle = \sum_{i=0}^{n-1} \alpha_i |j\rangle\langle j|i\rangle.$$

Further manipulations

This results in

$$\mathbf{P}_j|\psi\rangle = \alpha_j|j\rangle,$$

since $\langle j|i\rangle = \delta_{ji}$. With the last equation we can rewrite

$$\mathbf{A}|\psi\rangle = \sum_{i=0}^{n-1} \alpha_i \lambda_i |i\rangle = \sum_{i=0}^{n-1} \lambda_i \mathbf{P}_i |\psi\rangle,$$

from which we conclude that

$$\mathbf{A} = \sum_{i=0}^{n-1} \lambda_i \mathbf{P}_i.$$

Spectral decomposition

This is the spectral decomposition of a hermitian and normal operator. It is true for any state and it is independent of the basis. The spectral decomposition can in turn be used to exhaustively specify a measurement, as we will see in the next section.

As an example, consider two states $|\psi_a\rangle$ and $|\psi_b\rangle$ that are eigenstates of $\mathbf{A}$ with eigenvalues λ_a and λ_b , respectively. In the diagonalization process we have obtained the coefficients α_0 , α_1 , β_0 and β_1 using an expansion in terms of the orthogonal basis $|0\rangle$ and $|1\rangle$.

Explicit results

We have then

$$|\psi_a\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle,$$

and

$$|\psi_b\rangle = \beta_0|0\rangle + \beta_1|1\rangle,$$

with corresponding projection operators

$$\mathbf{P}_a = |\psi_a\rangle\langle\psi_a| = \begin{bmatrix} |\alpha_0|^2 & \alpha_0\alpha_1^* \\ \alpha_1\alpha_0^* & |\alpha_1|^2 \end{bmatrix},$$

and

$$\mathbf{P}_b = |\psi_b\rangle\langle\psi_b| = \begin{bmatrix} |\beta_0|^2 & \beta_0\beta_1^* \\ \beta_1\beta_0^* & |\beta_1|^2 \end{bmatrix}.$$

The spectral decomposition

The results from the previous slide gives us the following spectral decomposition of $\mathbf{A}$

$$\mathbf{A} = \lambda_a |\psi_a\rangle \langle \psi_a| + \lambda_b |\psi_b\rangle \langle \psi_b|,$$

which written out in all its details reads

$$\mathbf{A} = \lambda_a \begin{bmatrix} |\alpha_0|^2 & \alpha_0 \alpha_1^* \\ \alpha_1 \alpha_0^* & |\alpha_1|^2 \end{bmatrix} + \lambda_b \begin{bmatrix} |\beta_0|^2 & \beta_0 \beta_1^* \\ \beta_1 \beta_0^* & |\beta_1|^2 \end{bmatrix}.$$

First exercise set

The last two exercises are meant to build the basis for the two projects we will work on during the semester. The first project deals with implementing the so-called **Variational Quantum Eigensolver** algorithm for finding the eigenvalues and eigenvectors of selected Hamiltonians.

Show that the so-called Bell states listed here (and to be encountered many times in this course) form an orthogonal basis

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} [|00\rangle + |11\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} [|00\rangle - |11\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix},$$

And the next two

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} [|10\rangle + |01\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix},$$

and

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} [|10\rangle - |01\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}.$$

Show that the state $\alpha|00\rangle + \beta|11\rangle$ cannot be written as the product of the tensor product of two states and is thus entangled. The constants α and β are both nonzero.

Prove the following commutator relations for different operators (marked with a hat)

1. $[\hat{A} + \hat{B}, \hat{C}] = [\hat{A}, \hat{C}] + [\hat{B}, \hat{C}];$
2. $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}];$ and
3. $[\hat{A}, [\hat{B}\hat{C}]] = [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0$ (the so-called Jacobi identity).

1. Set up the commutation rules for Pauli matrices, that is find $[\sigma_i, \sigma_j]$ where $i, j = x, y, z$.
2. We define $\mathbf{X} = \sigma_x$, $\mathbf{Y} = \sigma_y$ and $\mathbf{Z} = \sigma_z$. Show that $\mathbf{XX} = \mathbf{YY} = \mathbf{ZZ} = I$.
3. Which one of the Pauli matrices has the qubit basis $|0\rangle$ and $|1\rangle$ as eigenbasis? What are the eigenvalues?

Prove that if two operators $\hat{A}$ and $\hat{B}$ commute they will share a basis of eigenstates

Write a function which sets up a one-qubit basis and apply the various Pauli matrices to these basis states.

Apply the Hadamard and Phase gates to the same one-qubit basis states and study their actions on these states.